TinyBolus Formulary

Peer-review document

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Developed by Sérgio Pinto, M.D.

Contact: tinybolusfeedback@gmail.com
Source: github.com/SGP-Prime/tinybolusformulas
Privacy policy: sgp-prime.github.io/tinybolus-site/privacy.html

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This is a reference tool, not a prescribing system. Verify every value against your hospital protocols, product labelling, and the individual patient in front of you before administration.

About this document

This is the human-readable export of the TinyBolus paediatric reference formulary, organised section-by-section as it appears in the application. Each item is presented with its dose specification, applicable age brackets (where present), route, clinical notes, and numerical references pointing into the bibliography at the end of the document.

How to read each item

Items fall into a small number of shapes that map directly onto how the application computes them:

- **Single dose** (e.g., 10 µg/kg): one factor multiplied by weight, often with a hard maximum cap.
- **Dose range** (e.g., 2–3 mg/kg): a low–high range, also weight-scaled, optionally with a maximum cap on either bound.
- **Age-bracketed**: different dose factors for different age cohorts. Rendered as a small table per item.
- **Age-bracketed fixed value**: not weight-scaled, just a string per age band (e.g., laryngoscope blade size).
- **Computed**: piecewise logic that doesn't reduce to a single formula (ETT sizing, fluid maintenance, LMA / supraglottic airway sizing, vital-sign normal ranges).

Hard floors / contraindications

Some items carry an *age* or *weight* floor below which the formulary explicitly declines to suggest a dose. In the app this renders as `∅` (empty-set glyph) with a "not indicated" explanation. Items with such floors are flagged in red below each row.

References

Every dose / range / age bracket / sizing rule carries one or more citations. References are numbered in encounter order throughout this document, with a full bibliography at the end. The numbers in [brackets] after each item refer to entries in that bibliography.

Source

The formulary itself lives at github.com/SGP-Prime/tinybolusformulas as a single machine-readable JSON file. This document is regenerated from that file on demand by `tool/generate_formulary_review_html.dart`; do not hand-edit it.

Contents

EMERGENCY — PALS

Compression rate

VALUE `100–120/min`

References: [1]

Compression depth

RULE `4 cm <1 y · 5 cm ≥1 y`

not exceeding 6 cm at any age

References: [1]

C:V ratio (2 rescuers)

VALUE `15:2`

1 rescuer: 30:2

References: [1]

Defibrillation

DOSE `4 J/kg`

escalate to 8 J/kg (max 360 J) for refractory VF/pVT from the 6th shock

References: [1], [2]

Cardioversion (synchronised)

DOSE `1 J/kg`

double per attempt to max 4 J/kg

References: [1]

Adrenaline (cardiac arrest)

IV/IO

DOSE `10 µg/kg (max 1 mg)`

repeat every 3–5 min

References: [1], [3]

Amiodarone

IV

DOSE `5 mg/kg (max 300 mg)`

max 150 mg for subsequent doses

References: [1]

Atropine

IV

≤11 y	20 µg/kg (max 500 µg)
≥12 y	20 µg/kg (max 1 mg)

20 µg/kg · max single dose 0.5 mg child / 1 mg adolescent · cumulative max 2 mg · no minimum (Barrington 2011)

References: [1], [4], [5]

Adenosine (1st dose)

IV

DOSE `0.1 mg/kg (max 6 mg)`

References: [1]

Adenosine (2nd dose)

IV

DOSE `0.3 mg/kg (max 18 mg)`

if SVT persists, escalate in 0.05–0.1 mg/kg steps to a max single dose of 0.5 mg/kg (ERC 2025)

References: [1]

Vital Signs

Heart Rate

≤1 mo	120–170 bpm
2–12 mo	100–160 bpm
1–2 y	98–140 bpm
3–5 y	80–120 bpm
6–11 y	75–118 bpm
≥12 y	60–100 bpm

References: [6]

Systolic BP

≤12 mo	72–104 mmHg
1–2 y	86–106 mmHg
3–5 y	89–112 mmHg
6–9 y	97–115 mmHg
10–11 y	102–120 mmHg
12–15 y	110–131 mmHg
≥16 y	90–140 mmHg

References: [6]

Diastolic BP

≤12 mo	37–56 mmHg
1–2 y	42–63 mmHg
3–5 y	46–72 mmHg
6–9 y	57–76 mmHg
10–11 y	61–80 mmHg
12–15 y	64–83 mmHg
≥16 y	60–90 mmHg

References: [6]

Respiratory Rate

≤1 mo	40–60 /min
2–12 mo	30–53 /min
1–3 y	22–37 /min
4–5 y	20–28 /min
6–12 y	18–25 /min
≥13 y	12–20 /min

References: [6]

Tidal Volume

DOSE `5–7 mL/kg`

intraoperative ventilation, healthy lungs · use lower end (5 mL/kg) for neonates and very young infants

References: [7], [8]

Induction

Propofol

IV

≤1 mo	1–2.5 mg/kg
2–12 mo	3–4 mg/kg
1–6 y	3–3.5 mg/kg
7–12 y	2.5–3 mg/kg
≥13 y	2–2.5 mg/kg

after a sevoflurane inhalational induction, roughly halve the dose (lower requirement)

References: [9], [10], [11], [12]

Ketamine

IV

DOSE `1–2 mg/kg`

References: [13], [14]

Ketamine (IM)

IM

DOSE `4–5 mg/kg`

induction/sedation when no IV access · onset 2–5 min

References: [13], [14]

Rocuronium

IV

DOSE `0.6–1.2 mg/kg`

intubation 0.6 · RSI 1.0–1.2

References: [15], [16]

Succinylcholine (RSI)

IV

≤12 mo 2–3 mg/kg — *raised in infants — larger extracellular-fluid volume of distribution (Meakin 1989)*

≥1 y 1–2 mg/kg

intubating dose · laryngospasm + IM doses in Respiratory

References: [16], [6], [17]

Fentanyl (induction)

IV

≤1 mo 1–3 µg/kg — *neonate: reduce dose · inject slowly (chest-wall rigidity) · monitor for apnoea (SPA 2019)*

≥0 y 1–3 µg/kg

References: [14], [18]

Midazolam

IV

≤6 mo 0.1 mg/kg — *young infant <6 mo: reduced clearance — prolonged effect · slow injection · hypotension, esp. with an opioid*

≥0 y 0.1 mg/kg

CAP `max 5 mg`

References: [14]

Midazolam (oral premed)

PO

DOSE `0.5 mg/kg (max 20 mg)`

References: [14]

Midazolam (buccal premed)

BUCCAL

DOSE `0.3 mg/kg (max 10 mg)`

buccal alternative to oral premed (e.g. swallowing refused) · onset ~10–15 min

References: [14]

Midazolam (IN premed)

IN

DOSE `0.2–0.3 mg/kg (max 10 mg)`

intranasal premedication · 0.3 mg/kg for faster, more reliable onset

References: [19], [14]

Ondansetron

IV

DOSE `0.1 mg/kg (max 4 mg)`

References: [20], [14]

Dexamethasone (PONV)

IV

DOSE `0.15–0.25 mg/kg (max 4 mg)`

PONV prophylaxis at induction · 0.25 mg/kg effective ceiling (no benefit above)

References: [20], [21], [14]

Dexamethasone (anti-inflammatory)

IV

DOSE `0.5 mg/kg (max 8 mg)`

airway oedema / post-extubation stridor · 0.5 mg/kg/dose (vs the 0.15 mg/kg PONV dose above)

References: [22], [23]

Etomidate

IV

DOSE `0.2–0.3 mg/kg`

haemodynamically stable induction

References: [14], [6]

Atracurium

IV

DOSE `0.4–0.5 mg/kg`

intubating dose · 0.3–0.4 mg/kg in infants

References: [14], [6]

Dexmedetomidine (IN premed)

IN

DOSE `1–3 μg/kg (max 200 μg)`

intranasal premedication · 2 μg/kg usual, up to 3 in older/under-sedated · onset 30–45 min

References: [24], [25]

Painkillers

Paracetamol

IV

≤10 kg 7.5 mg/kg

>10 kg 15 mg/kg

CAP `max 1 g`

IV dosing splits by weight, not age (SmPC): ≤10 kg → 7.5 mg/kg

References: [26], [14], [27]

Ketorolac

IV

DOSE `0.5 mg/kg (max 30 mg)`

Contraindicated <6 months

References: [28], [14]

Morphine

IV

≤1 mo 0.03–0.05 mg/kg

≥0 y 0.1 mg/kg

References: [29], [14]

Fentanyl (analgesia)

IV

≤1 mo 0.5–1 µg/kg — *neonate: reduce dose · inject slowly · monitor for apnoea (SPA 2019)*

≥0 y 0.5–1 µg/kg

References: [14], [30], [18]

Metamizole

IV

≤11 mo	5 mg/kg — *reduced in infants < 1 y — higher exposure per dose from immature metabolism (Ziesenitz 2019)*
≥0 y	10–20 mg/kg

CAP `max 1 g`

Contraindicated <3 months

References: [31], [32]

Ibuprofen (IV)

IV

DOSE `10 mg/kg (max 400 mg)`

References: [33], [14]

Ibuprofen (oral)

PO

DOSE `5–10 mg/kg (max 400 mg)`

analgesic / antipyretic · typically 7–10 mg/kg/dose

References: [33], [14]

Emergency Drugs

Adrenaline (cardiac arrest)

IV/IO

DOSE `10 µg/kg (max 1 mg)`

repeat every 3–5 min

References: [1], [3]

Adrenaline (anaphylaxis IM)

IM

DOSE `10 µg/kg (max 500 µg)`

weight-based 10 µg/kg (shown) · autoinjector equivalents: <6 y 150 µg · 6–12 y 300 µg · >12 y 500 µg (EAACI)

References: [34]

Adrenaline (anaphylaxis IV)

IV

DOSE `1 µg/kg`

titrate to response · IM adrenaline is first-line

References: [1], [34]

Atropine

IV

≤11 y	20 µg/kg (max 500 µg)
≥12 y	20 µg/kg (max 1 mg)

20 µg/kg · max single dose 0.5 mg child / 1 mg adolescent · cumulative max 2 mg · no minimum (Barrington 2011)

References: [1], [4], [5]

Ephedrine

IV

≤6 mo	0.6–1.2 mg/kg — *young infants need ~10× the adult dose (indirect agent · immature SNS) · 1.2 mg/kg optimal (Szostek 2023)*
≥0 y	0.1–0.3 mg/kg

References: [35], [14], [6]

Phenylephrine

IV

DOSE `5–20 µg/kg (max 500 µg)`

vasopressor — hypotension (also tet spells)

References: [14], [6]

Lidocaine

IV

DOSE `1 mg/kg (max 100 mg)`

antiarrhythmic — alternative to amiodarone for shock-refractory VF/pVT

References: [1], [3]

Glucose 10%

IV/IO

DOSE `2 mL/kg`

0.2 g/kg for hypoglycaemia · recheck glucose after

References: [1], [5]

Calcium gluconate 10%

IV/IO

DOSE `0.5 mL/kg (max 30 mL)`

hyperkalaemia · hypocalcaemia · CCB toxicity · give slowly

References: [1], [5]

Sodium bicarbonate 8.4%

IV/IO

DOSE `1 mmol/kg`

1 mL/kg of 8.4% · for hyperkalaemia or TCA toxicity, not routine arrest

References: [1], [5]

Dantrolene

IV

DOSE `2.5 mg/kg`

repeat 5–10 min · max 10 mg/kg

References: [36], [37]

Fluids and Blood

Fluid Bolus

IV

DOSE `10 mL/kg`

References: [1]

Maintenance

RULE `Holliday-Segar 4-2-1: 4 mL/kg/h (first 10 kg) · 2 mL/kg/h (next 10 kg) · 1 mL/kg/h thereafter`

4-2-1 rule (Holliday-Segar)

References: [38], [39]

Total Blood Volume

≤1 mo	85 mL/kg
2 mo – 2 y	75 mL/kg
≥3 y	70 mL/kg

References: [6]

Tranexamic Acid

IV

DOSE `15 mg/kg (max 1 g)`

loading dose · then 2 mg/kg/h infusion (RCPCH)

References: [40], [41], [42]

Equipment

ETT

RULE `Cuffed: 3.5 + age/4, rounded to ±0.5`

≤4 KG `Neonatal weight table:`

≤1 kg	size 2.5
≤2 kg	size 3.0
≤4 kg	size 3.5

INFANT `age table (cuffed):`

≤6 mo	size 3.5
7–12 mo	size 4.0

for uncuffed tube, add 0.5 to size

References: [43]

ETT insertion depth

RULE `6 + weight (<1 mo) · weight/2 + 8 (<1 y) · age/2 + 12 (≥1 y)` *(cm)*

nasal: add 2–3 cm

References: [44], [45], [5]

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iGel LMA

IGEL SIZING `weight bands (inclusive on both bounds):`

2–5 kg	size 1
5–12 kg	size 1.5
10–25 kg	size 2
25–35 kg	size 2.5
30–60 kg	size 3
50–90 kg	size 4
90–∞ kg	size 5

References: [46], [47]

AuraGain LMA

AURAGAIN SIZING `weight bands (inclusive on both bounds):`

2–5 kg	size 1
5–10 kg	size 1.5
10–20 kg	size 2
20–30 kg	size 2.5
30–50 kg	size 3
50–70 kg	size 4
70–100 kg	size 5
100–∞ kg	size 6

References: [48]

Laryngoscope blade

≤1 y	Miller 0–1
2–11 y	Macintosh 2
≥12 y	Macintosh 3

infants: straight (Miller) blade preferred to lift the floppy epiglottis directly · Macintosh 0–1 an accepted alternative

References: [49], [6]

Face mask

≤1 mo	Size 0
2–12 mo	Size 1
1–3 y	Size 2
4–8 y	Size 3
9–14 y	Size 4
≥15 y	Size 5

mask should cover nose and mouth from eyebrow line to chin cleft · anatomical fit is more important than estimated size

References: [6], [50]

Reservoir bag

≤6 mo	0.5 L
7 mo – 6 y	1 L
7–12 y	2 L
≥13 y	3 L

anaesthesia breathing-circuit bag · size follows weight: ~0.5 L neonate, 1 L <25-30 kg, 2 L 30-50 kg, 3 L >50 kg

References: [51]

Suction catheter

≤1 mo	6 Fr
2 mo – 1 y	8 Fr
2–8 y	10 Fr
≥9 y	12–14 Fr

roughly twice the ETT internal diameter (Fr)

References: [5], [6]

NG/OG tube

≤1 mo	6 Fr
2 mo – 1 y	8 Fr
2–8 y	10 Fr
≥9 y	12–14 Fr

References: [5], [6]

Respiratory

Salbutamol (inhaled)

INHALED

≤5 y 2–10 puffs

≥6 y 4–10 puffs

100 µg/puff · repeat incrementally as needed

References: [52], [53]

Salbutamol (nebulized)

NEBULIZED

≤5 y 2.5 mg

≥6 y 5 mg

repeat back-to-back or continuously per response

References: [52], [53]

Ipratropium (inhaled)

INHALED

≤5 y 4 puffs

≥6 y 4–8 puffs

20 µg/puff · repeat as needed

References: [54], [52]

Ipratropium (nebulized)

NEBULIZED

≤11 y 250 µg

≥12 y 500 µg

References: [14]

Magnesium Sulphate

IV

DOSE `25–50 mg/kg (max 2 g)`

References: [55], [14]

Clemastine

IV

DOSE `25 µg/kg (max 2 mg)`

References: [56], [57]

Adrenaline (nebulized)

NEBULIZED

DOSE `0.5 mg/kg (max 5 mg)`

References: [58], [14]

Succinylcholine (laryngospasm)

IV

DOSE `0.1–0.5 mg/kg`

References: [59]

Succinylcholine (IM)

IM

DOSE `4 mg/kg (max 200 mg)`

for laryngospasm or RSI without IV access · 4 mg/kg (max 200 mg)

References: [59], [60]

Rocuronium (laryngospasm)

IV

DOSE `0.1–0.2 mg/kg`

when succinylcholine is unavailable or contraindicated

References: [60]

Reversal Agents

Naloxone (titrated)

IV

DOSE 0.01 mg/kg

post-anaesthesia reversal — titrate to respiration · repeat every 2–3 min

References: [14], [1]

Naloxone (overdose)

IV/IO/IM

DOSE 0.1 mg/kg (max 2 mg)

suspected opioid overdose with apnoea or unresponsiveness

References: [1], [14]

Flumazenil

IV

DOSE 10 µg/kg (max 200 µg)

max bolus 200 µg · total max 1 mg

References: [14]

Sugammadex

IV

DOSE 2–4 mg/kg

2 mg/kg moderate block · 4 mg/kg deep block

References: [61], [62]

Sugammadex (profound block)

IV

DOSE 16 mg/kg

for CICO / immediate reversal of full intubating-dose rocuronium

References: [63], [64]

Neostigmine

IV

DOSE `0.04–0.07 mg/kg (max 5 mg)`

coadminister with atropine to block muscarinic effects · give slowly over 1–2 min after first signs of spontaneous recovery

References: [14], [6]

Atropine (with neostigmine)

IV

DOSE `20 µg/kg (max 500 µg)`

give together with or just before neostigmine to blunt bradycardia and bronchorrhoea

References: [14], [6]

Regional Anesthesia

Lidocaine (perineural)

PERINEURAL

≤5 mo maximum 3.5 mg/kg

≥0 y maximum 5 mg/kg

References: [65], [14]

Bupivacaine (perineural)

PERINEURAL

≤5 mo maximum 1.5–2 mg/kg — *infusion: 0.2 mg/kg/h*

≥0 y maximum 2–2.5 mg/kg — *infusion: 0.4 mg/kg/h*

References: [65], [66]

Ropivacaine (perineural)

PERINEURAL

≤5 mo maximum 2 mg/kg — *infusion: 0.2 mg/kg/h*

≥0 y maximum 3 mg/kg — *infusion: 0.4 mg/kg/h*

References: [67], [68], [66]

Intralipid 20% bolus

IV

DOSE `1.5 mL/kg`

max total 12 mL/kg

References: [65], [69]

Intralipid 20% infusion

IV

DOSE `0.25 mL/min/kg`

References: [65], [69]

Weight Estimation

Used when the user does not supply a measured weight. <1 month and >12 years always require user input.

1–12 mo (infant)

FORMULA `weight (kg) = (0.5 × months) + 4`

References: [5]

1–5 y (young child)

FORMULA `weight (kg) = (2 × age) + 8`

References: [5]

6–12 y (school-age)

FORMULA `weight (kg) = (3 × age) + 7`

References: [70], [5]

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Citations numbered in order of first encounter throughout this document.

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